Text-to-Cypher: An Agentic Approach

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Outline

Text-to-Query: The Challenge

- **Text-to-Query**: Translating natural language questions into formal database queries.
- **Task**: Bridging the gap between informal natural language and formal languages
- **Goal**: Retrieving structured data from unstructured intent
- **Text-to-Cypher**: A specialized subset for Property Graph Databases.
- **Complexity**:
  - Graph structures are complex to express.
  - Cypher requires "visual" thinking (ASCII art style patterns).

Complexities & Core Sub-Tasks

1. **Relevant Concepts Identification**: Extracting key and relevant concepts such as *entities*, *relationships*, and *properties*.
2. **Relevant Formalisms Discovery**: Finding appropriate formal constructs (node labels, relationship types) of the underlying symbolical framework.
3. **Concepts-Formalisms Linking**: Mapping identified concepts to discovered formal constructs (Entity/Relationship/Property Linking).
4. **Formalisms Framework Exploitation**: Generating syntactically and semantically correct queries (in Cypher in our case).

Complexity

All these sub-tasks are inherently probabilistic or possibilistic in nature.

Research Landscape: Overview

Existing literature splits into two main directions:

1. **Benchmark Development**: Creating high-quality datasets for training and evaluation.
2. **Text-to-Query Solutions**: Designing architectures (Algorithmic vs. Agentic) to solve the translation task.

Our Focus

We leverage existing benchmarks to propose a novel, zero-shot agentic solution for Text-to-Cypher.

Research Landscape: Datasets

- **neo4j/text2cypher-2024v1**[1] (Used):
  - **Size**: 44k+ instances (39k train, 4.8k test).
  - **Pros**: Access to 17+ live, publicly available Neo4j databases.
  - **Relevance**: Enables execution-based evaluation without local setup.
- **megagonlabs/cypherbench**[2] (Excluded):
  - **Size**: 10k instances (Wikidata derived).
  - **Cons**: Requires resource-intensive local deployment via Docker (64-128GB RAM).
  - **Relevance**: Too heavy for rapid experimentation.

[1] [Ozsoy et al.(2024)Ozsoy, Messallem, Besga, and Minneci]
[2] [Feng et al.(2024)Feng, Papicchio, and Rahman]

Incomplete Solutions: The Identification-Discovery-Linking Assumption

- Most works[3] assume **Identification**, **Discovery**, and **Linking** already solved providing the entire database schema without any *filtering* or *pruning*.
- This may lead to excessively large prompts that:
  - exceed the model's maximum context window.
  - pollute context with irrelevant information.
- How do systems performs Identification, Discovery, and Linking?
  - **Algorithmic** vs **Agentic** approaches.

[3] [D'Abramo et al.(2025)D'Abramo, Zugarini, and Torroni]
[Longwell et al.(2024)Longwell, Ali Akbar Alavi, Zarrinkalam, and Ensan]
[Feng et al.(2024)Feng, Papicchio, and Rahman]
[Ozsoy et al.(2024)Ozsoy, Messallem, Besga, and Minneci]

Algorithmic Approaches[4]

- **Pipeline**: Entity Linking → Schema Extraction → Generation.
- **Pros**: Deterministic, faster inference, easier to debug.
- **Cons**:
  - they rely on strong, domain-specific assumptions that may not generalize
  - they lack of robust disambiguation mechanisms with multiple candidates
  - Relationship and Property Linking are often ignored or oversimplified.

[4] [Yang et al.(2026)Yang, Li, Hu, Yu, and Lu]

Retrieval Architectures: Agentic Approaches[5]

- **Loop**: ReAct cycle (Thought → Action → Observation).
- **Pros**:
  - Flexible, adaptable to various domains and tasks.
  - Can handle ambiguity and multiple candidates through iterative refinement.
- **Cons**:
  - Ensuring Efficiency and Effectiveness at scale remains an open challenge.
  - Higher latency and potential for loops.

[5] [Walter and Bast(2025a), Walter and Bast(2025b)]

Similarity Techniques: The Trade-offs

Constraint: Effective retrieval of short schema terms (e.g., DataCenter, founded_at).

- **Lexical Matching** (Levenshtein, Jaro-Winkler, Prefix, etc.):
  - **Pros**: Very fast and effective for matching short terms/typos.
  - **Cons**: Semantic gaps (Synonyms, Polysemy, etc.).
- **Contextual Dense Embeddings** (BERT-based transformers):
  - **Pros**: Effective for sentence-level semantics.
  - **Cons**: Ineffective for very short de-contextualized terms + high compute cost.
- **FastText** (Subword Static Word Embeddings (S-SWEs)):
  - **Pros**: Fast lookup; captures semantics of words/subwords w/o overhead.
  - **Cons**: Large memory requirements for storing the vocabulary.
- **Model2Vec**[6] (Static Distilled Embeddings) - **Our Choice**:
  - **Pros**: Pros of FastText + memory efficiency

[6] [Tulkens and van Dongen(2024)]

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System Architecture

- **Framework**: LangChain & LangGraph.
- **Core Engine**: GPT-4.1 (Zero-Shot).
- **Indexing Strategy**:
  - **Vector Store**: FAISS with Model2Vec (potion-retrieval-32M[7]). Indexes Node Labels, Relationship Types, Property Keys.
  - **Schema Index**: Inverted index for structural lookups (e.g., node-property mappings).

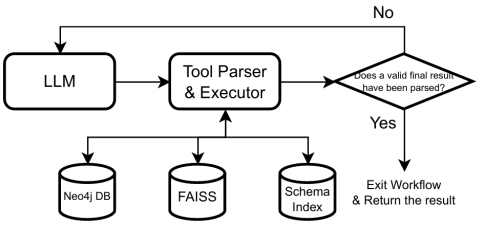


Figure: System Architecture

[7] which is a finetuned version of the potion-base-32M

Tool Design: Semantic Discovery

```python
def search_for(
    components: List[Literal["nodes", "relationships",
    ↪  "properties"]],
    similar_to: str,
    top_k: int = 10
) -> str:
```

- **Purpose**: Semantic search over indexed schema components (FAISS + Model2Vec).
- **Strategy**: Prioritizes *Recall* over Precision to cast a wide initial net.
- **Usage**: Invoked efficiently in parallel for independent concepts in the user query.

Tool Design: Property Retrieval

```python
def get_properties_from(
    components: List[Literal["nodes", "relationships"]],
    of_types: List[str],
    similar_to: Optional[List[str]] = None,
    top_k: Union[int, List[int]] = 10
) -> str:
```

- **Purpose**: Retrieves properties for explicit node labels or relationship types.
- **Batching**: Processes multiple components in a single call to reduce LLM turns.
- **Context**: Allows filtering property lists by semantic relevance (Model2Vec).

Tool Design: Structure Exploration

```python
def get_domains_ranges_of(
    relationships: List[str],
    similar_to: Optional[List[str]] = None,
    order_by: List[Literal["domains", "ranges"]] = None,
    top_k: Union[int, List[int]] = 10
) -> str:
```

- **Purpose**: Discovers valid Source (Domain) and Target (Range) nodes for relationships.
- **Graph Traversal**: essential for constructing valid path expressions.
- **Filtering**: Can prioritize certain domains/ranges based on semantic similarity to query concepts (Model2Vec).

Tool Design: Reverse Lookup

```python
def get_components_with_property(
    keys: List[str],
    components: List[Literal["nodes", "relationships", "both"]],
    similar_to: Optional[List[str]] = None,
    top_k: Union[int, List[int]] = 10
) -> str:
```

- **Purpose**: Finds schema components that possess specific property keys.
- **Disambiguation**: Uses discriminative properties (e.g., `created_at`, `isbn`) to filter candidate types.
- **Mechanism**: Leveraging the inverted schema index for efficient reverse lookups.

Tool Design: Grounding & Validation

```python
def execute_cypher_query(cypher: str) -> List[dict]:
```

- **Purpose**: Executes arbitrary Cypher queries against the live database.
- **Grounding**: Validates schema assumptions against actual data instances.
- **Data Exploration**: Checks for specific patterns, value formats, or edge cases.

Tool Design: Result Registration

```python
def register_retrieval_result(
    motivation: str,
    cypher_query: Optional[str] = None,
    kg_schema: Optional[KGSchema] = None
) -> RetrievalAgent.Result:
```

- **Purpose**: Signals the completion of the ReAct loop.
- **Output**: Returns specific structured data (Motivation, Query, Schema Subgraph).
- **Integration**: Passes the final result to the downstream execution system.

Experimental Setup

- **Dataset**: Subset of text2cypher-2024v1 (2,471 instances with live DBs).
- **Metrics**:
  - **Result Similarity (RS)**: Jaccard similarity of result sets.
  - **Exact Match (EM)**: RS $= 1.0$.
  - **Success Rate**: Completion without exceptions or loops.
  - **Provenance Subgraph Jaccard Similarity (PSJS)**: Jaccard similarity of retrieved schema subgraph vs gold schema subgraph.
- **Baselines**: Comparison with Zero-shot and Fine-tuned GPT-4o (from Neo4j Labs paper).

Effectiveness Results

Comparison with best baselines of Neo4j-Labs Team[8].
Our zero-shot agent matches fine-tuned performance.

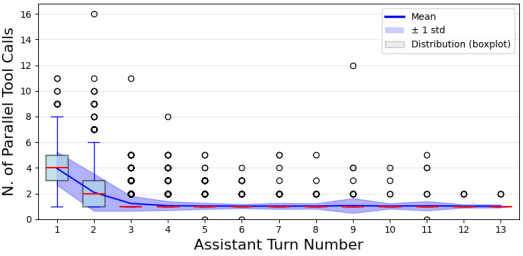
Model	Result Sim.	Exact Match	PSJS	Success Rate
GPT-4o (zero-shot)	-	31.73%	-	-
GPT-4o (fine-tuned)	-	32.50%	-	-
GPT-4.1 (Ours)	**41.22%**	**32.62%**	**69.37%**	89.11%

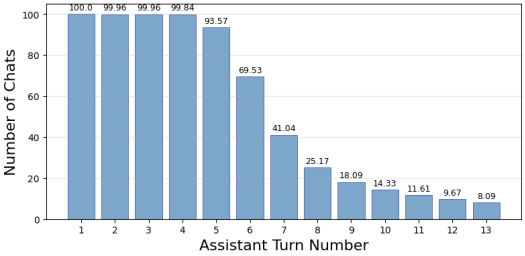
[8][Ozsoy et al.(2024)Ozsoy, Messallem, Besga, and Minneci]

Efficiency Analysis: Absolute Tool Parallelism

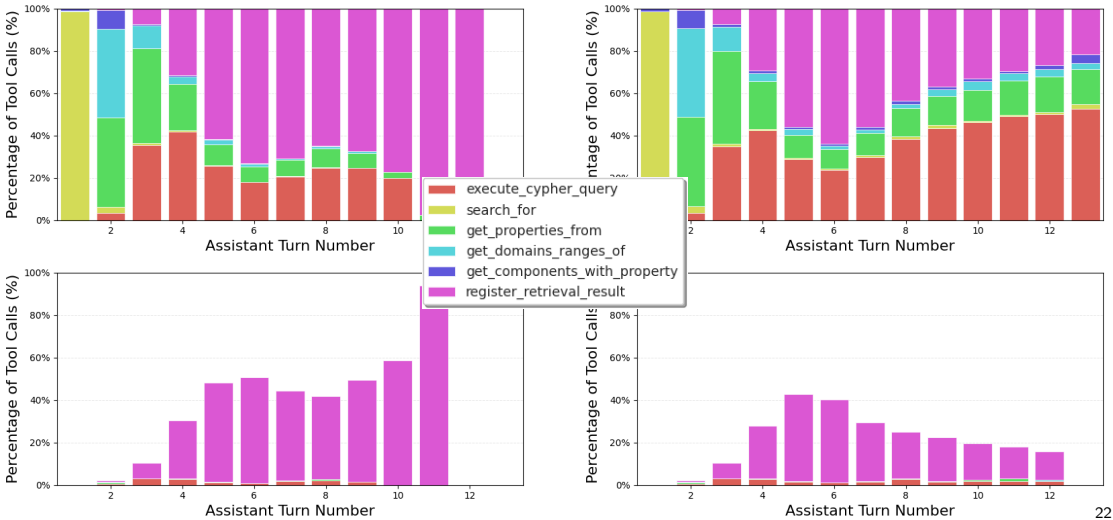
Tool	Mean	Std	Mode	Min	Median	Max	Count
execute_cypher_query	1.01	0.10	1	1	1	4	4342
search_for	3.69	1.43	4	1	4	11	2614
get_properties_from	1.42	0.76	1	1	1	16	3813
get_domains_ranges_of	1.46	0.90	1	0	1	12	1924
get_components_with_property	2.05	1.47	1	1	2	9	523
register_retrieval_result	1.00	0.00	1	1	1	1	4659

Efficiency Analysis: Turn Parallelism and Distribution





Efficiency Analysis: Tool Calls and Errors Distributions



Conclusions

- **Agentic Approach**: Successfully bridges NL and formal queries without fine-tuning.
- **Performance**: Matches fine-tuned baselines (EM $\approx 32.6\%$).
- **Strategy**: Validated "search-then-refine" behavior with high initial parallelism.
- **Limitations**:
  - Struggle with complex, non-linear reasoning paths (backtracking).
  - JSON formatting errors in final result registration.
- **Future Work**:
  - Robust error recovery mechanisms.
  - Hybrid Fine-tuning for tool usage and formatting.

Thank You!

Questions?

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https://github.com/ashkihotah

M. G. Ozsoy, L. Messallem, J. Besga, G. Minneci, Text2cypher: Bridging natural language and graph databases, 2024. URL: https://arxiv.org/abs/2412.10064. arXiv:2412.10064.

Y. Feng, S. Papicchio, S. Rahman,
Cypherbench: Towards precise retrieval over full-scale modern knowledge graphs in the llm era,
arXiv preprint arXiv:2412.18702 (2024).

J. D'Abramo, A. Zugarini, P. Torroni,
Investigating large language models for text-to-SPARQL generation,
in: W. Shi, W. Yu, A. Asai, M. Jiang, G. Durrett, H. Hajishirzi, L. Zettlemoyer (Eds.), Proceedings of the 4th International Workshop on Knowledge-Augmented Methods for Natural Language Processing, Association for Computational Linguistics, Albuquerque, New Mexico, USA, 2025, pp. 66–80. URL:

https://aclanthology.org/2025.knowledgenlp-1.5/.
doi:10.18653/v1/2025.knowledgenlp-1.5.


J. Longwell, M. Ali Akbar Alavi, F. Zarrinkalam, F. Ensan,
Triple augmented generative language models for sparql query generation from natural language questions,
in: Proceedings of the 2024 Annual International ACM SIGIR Conference on Research and Development in Information Retrieval in the Asia Pacific Region, SIGIR-AP 2024, Association for Computing Machinery, New York, NY, USA, 2024, p. 269–273. URL: https://doi.org/10.1145/3673791.3698426.
doi:10.1145/3673791.3698426.


C. Yang, C. Li, X. Hu, H. Yu, J. Lu,
Enhancing knowledge graph interactions: A comprehensive text-to-cypher pipeline with large language models,

Information Processing & Management 63 (2026) 104280. URL: https://www.sciencedirect.com/science/article/pii/S0306457325002213. doi:https://doi.org/10.1016/j.ipm.2025.104280.


S. Walter, H. Bast,
GRASP: generic reasoning and SPARQL generation across knowledge graphs,
in: ISWC (1), volume 16140 of *Lecture Notes in Computer Science*, Springer, 2025a, pp. 271–289.


S. Walter, H. Bast,
Knowledge graph entity linking via interactive reasoning and exploration with GRASP,
in: OM 2025 (Ontology Matching Workshop), CEUR Workshop Proceedings, CEUR-WS.org, 2025b. To appear.


S. Tulkens, T. van Dongen, Model2vec: Fast state-of-the-art static embeddings, 2024. URL: https://github.com/MinishLab/model2vec. doi:10.5281/zenodo.17270888.